Time Series Regression and Anomalies of Errors

These notes accompany Lecture 10. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Lectures 8 and 9 modelled a single series using only its own past. Here we return to regression – but with time series data, which brings assumptions and failures of its own. The running example is a model of Polish inflation, built up in five steps.

Learning outcomes – after this lecture you should be able to:

1. Distinguish static, finite distributed lag and autoregressive distributed lag models, and compute impact and long-run multipliers.
2. Use level-shift and slope-shift dummies to handle structural breaks, and test for breaks with Bai-Perron.
3. State the six time series assumptions TS.1–TS.6 and distinguish strict from weak exogeneity.
4. Say precisely what breaks when each assumption fails – separating bias from invalid inference.
5. Test residuals for autocorrelation, heteroskedasticity and non-normality.
6. Choose an appropriate remedy for autocorrelated or heteroskedastic errors, and explain what each remedy assumes.

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1. Time series regression models

Three model types are useful in empirical time series work and all are easily estimated by OLS. We develop them through one example: the determinants of inflation in Poland.

1.1 The economic setting

Theory links inflation (cpi_t^{PL}) to the degree of slack in the economy, measured by the output gap ($\widehat{gdp}_t$). Polish inflation should also depend on inflation in its main trading partner, the euro area (cpi_t^{ea}), and on the zloty exchange rate (xr_t^{eurpln}) – the latter capturing imported inflation. The real exchange rate gap ($\widehat{rer}_t^{eurpln}$) is added to account for changes in importers' marginal costs.

> **Before you estimate anything**
>
> The variables must be stationary, or cointegrated. This is not optional – Lecture 9 showed that regressions on non-stationary series can be entirely spurious.
>
> Here we take logarithms to linearise trends, then difference the series. The output gap and the real exchange rate gap are stationary by construction (deviations of real GDP and REER from their HP-filtered trends).

1.2 Static models

A static model is postulated when changes in explanatory variables (z) at time t are believed to have an immediate effect on y. All variables are dated contemporaneously:

$$\Delta lncpi_t^{pl} = \beta_0 + \beta_1 \widehat{gdp}_t + \beta_2 \widehat{rer}_t^{eurpln} + \beta_3 \Delta lncpi_t^{ea} + \beta_4 \Delta lnxr_t^{eurpln} + u_t$$

So, in general, $\Delta y_t = \beta \Delta z_t$ when $\Delta u_t = 0$. Estimated on the data, this is not a good model: R^2 is low by time series standards and most coefficients are insignificant.

The likely reason is misspecification. Changes in economic conditions influence the behaviour of economic agents with a lag, not contemporaneously. For example, a change in the exchange rate may pass into prices only when the next batch of imported goods reaches the market.

1.3 Finite distributed lag models

In a finite distributed lag (FDL) model we allow one or more variables to affect y with a lag:

$$\Delta ln\ cpi_t^{PL} = \alpha_0 + \delta_1\, \widehat{gdp}_{t-2} + \delta_2\, \widehat{rer}_{t-1}^{eurpln} + \delta_3\, \Delta ln\ cpi_t^{EA} + \delta_4\, \Delta ln\ cpi_{t-1}^{EA} + \delta_5\, \Delta lnxr_{t-1}^{eurpln} + u_t$$

Coefficient	Interpretation
δ_3	The immediate change in Polish inflation from a 1 p.p. increase in EA inflation at time t – the impact multiplier
δ_4	The change one period after a temporary change in EA inflation
$\delta_3 + \delta_4$	The change one period after a permanent change in EA inflation – the long-run multiplier

The sum of the coefficients on the current and lagged explanatory variable is the long-run change in y given a permanent increase in that variable. Here there are no further changes after one period.

Estimated, the coefficients are now significant and correctly signed. But the model is still misspecified: it does not account for persistence in inflation itself.

1.4 Autoregressive distributed lag models

Prices are sticky, so inflation is persistent – shocks pass through gradually rather than dying out at once. Some firms cannot adjust prices immediately because of long-term contracts with suppliers and workers; adjusting prices is itself costly because of menu costs. To capture this we add lags of the dependent variable:

$$\Delta ln\ cpi_t^{PL} = \mu_0 + \gamma_1 \Delta ln\ cpi_{t-1}^{PL} + \gamma_2\ \widehat{gdp}_{t-2} + \ldots + \gamma_6\ \Delta lnxr_{t-1}^{eurpln} + u_t$$

Coefficient	Interpretation
γ_1	The persistence of inflation
γ_6	The short-run exchange rate pass-through
$\gamma_6 / (1 - \gamma_1)$	The long-run exchange rate pass-through – the effect of a permanent change in the exchange rate

Why persistence matters for policy

If γ_1 is high, shocks have a very long influence on inflation. Restoring equilibrium then requires a large policy response – the central bank must raise the rate substantially and hold it there to return inflation to target.

This is why the coefficient on the lagged dependent variable is not a technical detail. It is the parameter that tells you how expensive disinflation will be.

2. Dummy variables and structural breaks

The ADL results look economically reasonable – but adequacy is judged on the residuals, not only on the coefficients.

2.1 A break in the intercept

The residuals of the ADL model are mostly positive and correlated after 2020. In 2020 both the Polish and euro area economies were in recession, and the model predicts deflation because of the large negative output gap. Actual Polish inflation did not fall as much as predicted – possibly because producers are reluctant to cut prices.

Whatever the reason, unobserved factors pushed inflation up persistently over that period, and good practice is to account for them with a dummy:

$$D_t^{2020s} = 1 \text{ for } t = Q1\ 2020 - Q4\ 2025, \quad 0 \text{ otherwise}$$

Adding $\gamma_7 D^{2020s}$ to the ADL model gives a coefficient of about 0.47, significant at better than 0.1% (t = 3.62). Holding everything else fixed – past inflation, EA pass-through, the output gap, the real exchange rate gap and depreciation – baseline quarterly inflation has been about 0.47 p.p. higher since the COVID-era break.

This is a permanent shift, not a temporary spike

Plausible structural explanations: global supply-chain disruptions becoming structural rather than transitory (semiconductor shortages, shipping cost spikes); post-pandemic labour shortages, partly

from emigration and partly from sectoral mismatch, pushing wage growth up through the wage-price channel; and a de-anchoring of inflation expectations after several years of inflation running above target.

The econometrics identifies the break. Which of these explanations is right is an economic question the dummy cannot answer.

2.2 A break in the slope

The residuals of the augmented model reveal a second pattern: in 2000–2001 they are consistently positive and correlated, in 2002–2003 consistently negative. Two institutional facts bear on this – the monetary policy framework changed to inflation targeting in 2004, and Poland joined the EU in May 2004. It is plausible that during the transition period inflation responded to aggregate demand differently.

To test this we use a dummy for 2000–2003, but interacted with the output gap rather than entered alone:

$$\ldots + \gamma_7 D_t^{2020s} + \gamma_8 D_t^{2003} \cdot \widehat{gdp}_{t-2} + u_t$$

Coefficient	Estimate	Meaning
γ_2	0.078	Immediate impact of the output gap on inflation outside 2000–2003 (after two quarters)
γ_8	0.408	How much stronger that impact was during 2000–2003
$\gamma_2 + \gamma_8$	≈ 0.486	Total impact during 2000–2003

So inflation was far more sensitive to demand pressure in the early period. γ_8 captures a structural change in the slope of the Phillips curve.

Why the Phillips curve flattened

The transition to inflation targeting and the accumulation of central bank credibility anchored inflation expectations better, which flattens the curve.

EU accession increased competition and made supply chains more flexible, so producers could no longer respond to higher demand through prices alone – more of the adjustment came through margins.

Note the general lesson: a dummy on its own shifts the intercept; a dummy interacted with a regressor shifts that regressor's slope. Which you need depends on what the economic intuition (and residuals) are telling you.

2.3 Testing formally: the Bai-Perron test

Guessing a break date is not good practice. The Bai-Perron test checks whether a regression's coefficients have shifted at some unknown point (or points) in the sample, letting the data find the dates that best split the sample into distinct regimes.

It works sequentially: it first asks whether allowing one break improves the fit enough to justify the extra parameters (0 versus 1 break), and if so keeps testing whether one more break adds significant

explanatory power (1 versus 2, and so on), stopping as soon as an additional break no longer clears the bar. This guards against over-fitting spurious breaks.

Testing the intercept only – the procedure identifies a break in Q1 2020, confirming the judgement made from the residual plot. Testing the break in the slope – the procedure identifies a break in Q4 2003.

2.4 Why breaks matter

A break is essentially a shift in a parameter value. Breaks in the estimation sample cause:

- The estimated parameters become a compromise between two regimes, so the forecast path is pulled toward whichever regime dominates the sample.
- The interval then fails in both directions at once: the unmodelled shift inflates the residual variance and widens the band, while the band stays centred on a biased path. Too wide and too rarely right.

The practical sequence: look at the data and the residuals first, then apply formal tests – but combine them with economic reasoning. You should be able to justify why a break exists (economic interpretation), not merely observe that a test found one.

3. OLS assumptions for time series regression

For diagnostics it is convenient to write the model in matrix form, with X collecting all independent variables for all time periods – an array with T rows and k columns:

$$Y = X\beta + u$$

$Y = (y_1, \dots, y_T)'$

$\beta = (\beta_0, \beta_1, \dots, \beta_k)'$

$u = (u_1, \dots, u_T)'$

$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{k1} \\ \dots & \dots & \dots & \dots \\ 1 & x_{1T} & \dots & x_{kT} \end{pmatrix}$$

3.1 Assumptions TS.1 – TS.3

Assumption	Statement
TS.1 Linear in parameters	The time series process follows a model linear in its parameters
TS.2 No perfect collinearity	No independent variable is constant nor a perfect linear combination of the others. Correlation among regressors is allowed; perfect correlation is not
TS.3 Zero conditional mean	$E(u_t \mid X) = 0$ for each t – the error at time t is uncorrelated with each explanatory variable in every time period

When TS.3 holds we say the explanatory variables are **strictly exogenous**. Under TS.1–TS.3 the OLS estimators are unbiased conditional on X, and therefore unconditionally as well.

3.2 Strict versus weak exogeneity

Strict exogeneity is demanding, and often unrealistic. It requires that the regressors cannot react to what has happened to y in the past – because if they do, u_t will be correlated with x_{t+s}.

> **Why this fails in macroeconomics. An example**
>
> The exchange rate is affected by past inflation, and by expected future inflation, because the central bank reacts to inflation by moving interest rates.
>
> So the regressor responds to the history of the dependent variable. Strict exogeneity is violated almost by construction in any macro model with a policy-sensitive regressor.

In practice we need only contemporaneous exogeneity – the regressors at time t must be uncorrelated with the error at time t, not in every period:

$$E(u_t \mid x_{t1}, \ldots, x_{tk}) = E(u_t \mid x_t) = 0$$

Such regressors are called weakly exogenous. This is sufficient for consistency of the OLS estimator – but unbiasedness still requires strict exogeneity.

3.3 Assumptions TS.4 – TS.6

Assumption	Statement	What it delivers
TS.4 Homoskedasticity	$\mathrm{Var}(u_t \mid X) = \sigma^2$ for all t	Together with TS.1–TS.3 and TS.5: OLS is BLUE
TS.5 No serial correlation	$\mathrm{Corr}(u_t, u_s \mid X) = 0$ for $t \neq s$	As above
TS.6 Normality	u_t independent of X and i.i.d. Normal(0, σ^2)	Exact t and F distributions, and valid confidence intervals, at any sample size

> **What "BLUE" actually means**
>
> Best – smallest variance among the estimators in the class.
>
> Linear – the estimator is a linear function of the dependent variable y. Note that this is about the estimator, not about the model: "linear in parameters" is assumption TS.1, which is a different statement.
>
> Unbiased – $E(\hat{\beta}) = \beta$.
>
> Estimator.

4. Model properties when assumptions fail

TS.1 holds by construction and TS.2 is easy to ensure. The question is what happens when TS.3 through TS.6 fail.

4.1 A regressor is correlated with the error (TS.3 fails)

$$y_t = \beta_1 x_t + \beta_2 z_t + u_t, \quad \text{with } E(z_t u_t) \neq 0$$

- Both $\hat{\beta}_1$ and $\hat{\beta}_2$ are in general biased.
- $\hat{\beta}_1$ is unbiased only if z_t is uncorrelated with x_t.

The simulation confirms it. With a true data generating process (DGP) of y = 0.75x + 0.60z + u where z = 0.5z(−1) + 1.5u + nrnd, estimating y on x and z recovers a coefficient on z well away from 0.60, because z is correlated with the unobserved u.

4.2 Autocorrelated errors, no lagged dependent variable (TS.5 fails)

$$y_t = \beta_1 x_t + u_t, \quad u_t = \rho u_{t-1} + v_t$$

- $\hat{\beta}_1$ is still unbiased – the estimator remains centred on the true value.
- But the estimator of the coefficient standard error is no longer consistent, so t-tests are unreliable.

In the simulation with y = 0.75z + u and both z and u following AR(0.9) processes, the estimated coefficient sits close to 0.75 while its standard error cannot be trusted.

4.3 Autocorrelated errors with a lagged dependent variable

$$y_t = \beta_1 x_t + \beta_2 y_{t-1} + u_t, \quad u_t = \rho u_{t-1} + v_t$$

Now the coefficients are biased. The chain of reasoning is worth following:

> **Why the lagged dependent variable is fatal here**
>
> y_t is correlated with u_t by construction – it is built from it.
>
> So y_{t-1} is correlated with u_{t-1}.
>
> But u_t depends on u_{t-1} through the AR(1) error process.
>
> Therefore y_{t-1} is correlated with u_t – which is exactly a violation of TS.3, with the regressor now being the lagged dependent variable itself.

The simulation bears this out: with a true DGP of y = 0.5y(−1) + 0.75z + u and AR(0.9) errors, both estimated coefficients are noticeably off.

4.4 Heteroskedastic errors (TS.4 fails)

- $\hat{\beta}$ is still unbiased.
- But it is no longer efficient – it no longer has the smallest variance.
- The standard errors cannot be used in inference, so t-statistics are unreliable.

4.5 Non-normal errors (TS.6 fails)

- In small samples, some test statistics – t and F among them – do not have their standard distributions.
- Consistency is unaffected, and inference remains available in large samples because the central limit theorem supplies limiting distributions for the test statistics – normal for the t-ratio, χ^2 for the Wald.

4.6 Summary of consequences

Failure	Coefficients	Standard errors	Severity
TS.3 – regressor correlated with u	Biased and inconsistent	Invalid	Serious – no sample size helps
TS.5 – autocorrelation, no lagged y	Unbiased	Inconsistent	Inference only
TS.5 – autocorrelation with lagged y	Biased	Invalid	Serious – reduces to a TS.3 failure
TS.4 – heteroskedasticity	Unbiased, but inefficient	Invalid	Inference and efficiency
TS.6 – non-normality	Unbiased and consistent	Valid asymptotically	Small samples only

The single most important line in this table is the third. Autocorrelation is a mild problem on its own – but combined with a lagged dependent variable, which almost every macro model contains, it becomes a bias problem.

5. Regression models diagnostics

5.1 Testing residuals for autocorrelation

The portmanteau Q-test

Estimate the model, recover the residuals, and compute the residual autocorrelation coefficients r_i = corr($\hat{u}_t$, $\hat{u}_{t-i}$) for i = 1, …, m. The test checks whether they are jointly zero:

$$H_0: r_1 = r_2 = \ldots = r_m = 0 \quad \textit{(no autocorrelation up to order m)}$$

In EViews: View → Residual Diagnostics → Correlogram-Q-statistics, then specify the lag length. The critical value is not reported but the p-value is. For the FDL model (Model 2) the p-values are below 0.05 at every lag, so we reject the null confidently: the residuals are autocorrelated.

The Breusch-Godfrey LM test

Regress the residuals on the original regressors and on lagged residuals – the auxiliary regression:

$$\hat{u}_t = \gamma_0 + \gamma_1 x_{1t} + \ldots + \gamma_k x_{kt} + \psi_1 \hat{u}_{t-1} + \ldots + \psi_m \hat{u}_{t-m} + v_t$$

If the residuals contain no systematic information they are not correlated, so the ψ coefficients should all be insignificant. Hence H_0: $\psi_1 = \ldots = \psi_m = 0$.

In EViews: View → Residual Diagnostics → Serial Correlation LM Test, specifying the lag length (4 is typical for quarterly data). The statistic is the product of the number of observations and the auxiliary R^2. For Model 2 the p-value is 0.0001, so the null of no autocorrelation is rejected at any reasonable level.

5.2 Detecting heteroskedasticity

The Q-test works the same way but on squared residuals: compute $r_i = \text{corr}(\hat{u}_t^2, \hat{u}_{t-i}^2)$ and test whether they are jointly zero. In EViews: View → Residual Diagnostics → Correlogram Squared Residuals. For the Poland model the p-values exceed conventional levels at all lags, so we fail to reject homoskedasticity.

Two LM tests, both familiar from Lecture 6, give the same verdict:

Test	Null hypothesis	p-value	Decision
Breusch-Pagan-Godfrey	Homoskedasticity	0.2285	Fail to reject
White	Homoskedasticity	0.0925	Fail to reject

For both, the rejection rule is: reject the null when the LM-statistic exceeds its critical value, or equivalently when the p-value falls below the chosen significance level.

5.3 Testing for normality

Use the Jarque-Bera statistic. Normality means no skewness – close to zero – and no excess kurtosis, so kurtosis close to three; the statistic would then be small and insignificant. In EViews: View → Residual Diagnostics → Histogram-Normality Test.

Model	p-value	Decision
Model 1 (static)	0.00	Reject normality
Model 3 (ADL)	0.237	Fail to reject – residuals show almost no skewness or excess kurtosis

Note that the better-specified model also produces better-behaved residuals. That is the usual pattern: normality failures are frequently a symptom of misspecification rather than a separate problem.

6. Dealing with error irregularities

6.1 Autocorrelation, case 1: no lagged dependent variable

Option A – add lags

The general remedy is to add lags of y and of the regressors. In the Poland example, the FDL model (Model 2) has clearly autocorrelated residuals and the null of no serial correlation is rejected. Adding the first lag of the dependent variable – moving to the ADL model (Model 3) – means the null can no longer be rejected.

> **An important warning about forecasting**
>
> Without a lagged dependent variable, OLS is unbiased but its standard errors are wrong. If your aim is inference, that is the whole problem.
>
> If your aim is forecasting, the damage is larger: leaving autocorrelated errors unmodelled means throwing away the predictable component of the error, so the forecast path itself is no longer the best available – not just the uncertainty around it.

Option B – add AR terms

Alternatively, add AR terms to the model and estimate by maximum likelihood. In principle up to m terms may be needed; here only AR(1) is required. But the two corrections are not interchangeable:

	ADL – lagged dependent variable	AR(1) error term
Where persistence sits	Explicitly modelled – structural persistence from sticky prices	Pushed into the error term – shocks are persistent, but there is no built-in inflation inertia
Economic reading	Price rigidity: long-term contracts, menu costs, asymmetric information. Response to changed conditions is stretched over several periods	Prices are flexible and respond quickly; only the shocks themselves are persistent
Policy implication	Disinflation is costly and slow; monetary policy must be forward-looking	Expectations are well anchored; policy need not be very restrictive, as disinflation is cheap

The two models can fit almost equally well and imply opposite policy advice. Choosing between them is an economic judgement, not a statistical one.

Option C – HAC standard errors

If the aim is only to check the significance of a regression, with no forecasting involved, use a heteroskedasticity and autocorrelation consistent estimate of the error variance. In EViews: Estimate → Options → Covariance method: HAC (Newey-West), with HAC Options → Automatic lag selection by Akaike. This corrects the standard errors and hence the t-statistics, leaving the coefficients unchanged.

6.2 Autocorrelation, case 2: with a lagged dependent variable

Here HAC will not save you, because the problem is bias rather than inference. The remedy is to include longer lags – in general y_{t-2} and z_{t-1}:

$$y_t = \beta_1 y_{t-1} + \beta_2 z_t + u_t \quad \Rightarrow \quad y_t = \alpha_1 y_{t-1} + \alpha_2 z_t + \alpha_3 y_{t-2} + \alpha_4 z_{t-1} + u_t$$

Consistent estimates of ρ and the β's can then be recovered from the relations:

$$\begin{cases} \beta_1 + \rho = \alpha_1 \\ -\beta_1 * \rho = \alpha_3 \\ \beta_2 = \alpha_2 \end{cases}$$

These are the common factor restrictions.

6.3 Heteroskedasticity

Heteroskedasticity says nothing about the conditional mean, so $\hat{\beta}$ survives unbiased and consistent. What fails is the variance formula, which assumes a constant σ^2 that the data do not have. Used for forecasting, the model produces an unbiased mean forecast but biased confidence intervals – the projection is right, the uncertainty around it is not.

Remedy	When to use
Model the variance explicitly and estimate by maximum likelihood – for example as an ARCH or GARCH process	When the variance itself is of interest, or when forecast uncertainty matters. Covered in Lecture 11
White (Huber-White) robust standard errors	When errors are heteroskedastic but not autocorrelated, and you only need valid inference
Newey-West (HAC) standard errors	When errors are both heteroskedastic and autocorrelated

In EViews: Estimate → Options, then select Huber-White or HAC under the covariance method.

7. Summary

1. Static, FDL and ADL models differ in how they allow time to enter. The ADL is usually the workhorse for macro data, because it captures persistence directly.
2. In an ADL model, the coefficient on the lagged dependent variable measures persistence, and the long-run multiplier divides the short-run effect by one minus that coefficient.
3. Structural breaks are handled with dummies: alone for a level shift, interacted with a regressor for a slope shift. Bai-Perron finds break dates from the data, but economic justification is still required.
4. No regression result should be reported before its residuals have been examined for autocorrelation and heteroskedasticity.
5. Autocorrelation costs you the standard errors always, and the coefficients as well whenever a lagged dependent variable sits on the right-hand side. Forecasts suffer twice over: the path is wrong, and so is the stated uncertainty around it.
6. Changing error variance leaves the central projection intact and corrupts the interval around it.
7. Remedies exist, and the choice among them follows from the purpose: robust standard errors are enough to restore valid tests, while only a change of specification restores unbiased coefficients and honest forecast intervals.

> **The diagnostic habit worth forming**
>
> The Poland example is a sequence of five models, and every step came from looking at the residuals – not from theory, and not from a test statistic alone.
>
> Residuals mostly positive after 2020 → level-shift dummy. Residuals positive then negative in 2000–2003 → slope-shift dummy. Residuals autocorrelated → add a lag.
>
> The tests confirm what the plot suggests. They rarely discover it for you.

Key terms

Term	Meaning
Static model	All variables dated contemporaneously; changes in z have an immediate effect on y.
Finite distributed lag (FDL)	One or more explanatory variables enter with lags.

Term	Meaning
Impact multiplier	The immediate effect of a one-unit change in a regressor.
Long-run multiplier	The total effect of a permanent change; the sum of current and lagged coefficients.
Autoregressive distributed lag (ADL)	Lags of the dependent variable appear among the regressors.
Persistence	The coefficient on the lagged dependent variable; governs how long shocks influence y.
Level-shift dummy	A dummy entered alone, shifting the intercept over a period.
Slope-shift dummy	A dummy interacted with a regressor, changing that regressor's coefficient over a period.
Bai-Perron test	A sequential test for structural breaks at unknown dates.
Strict exogeneity (TS.3)	$E(u_t \mid X) = 0$ – the error is uncorrelated with the regressors in every period. Needed for unbiasedness.
Weak (contemporaneous) exogeneity	$E(u_t \mid x_t) = 0$. Sufficient for consistency, not for unbiasedness.
Portmanteau Q-test	Joint test that residual autocorrelations up to lag m are zero.
Breusch-Godfrey LM test	Regresses residuals on the regressors and lagged residuals; tests the lagged-residual coefficients jointly.
HAC standard errors	Newey-West; valid under both heteroskedasticity and autocorrelation. Corrects standard errors, not coefficients.

Review questions

Use these to check your understanding before the seminar.

1. Why must the variables be stationary or cointegrated before estimating any of these models? Refer back to Lecture 9.
2. Distinguish the impact multiplier from the long-run multiplier in an FDL model. What does each answer?
3. In the ADL model, $\gamma_6 = 0.15$ and $\gamma_1 = 0.6$. Compute the long-run exchange rate pass-through and explain the formula.
4. Explain why a high value of γ_1 means disinflation will be expensive.
5. The residuals of Model 3 are mostly positive after 2020. Why is a dummy the right response, and what does its coefficient of 0.47 mean in words?
6. When would you enter a dummy alone, and when interacted with a regressor? Give the diagnostic signal for each.
7. $\gamma_2 = 0.078$ and $\gamma_8 = 0.408$. State the output-gap effect inside and outside 2000–2003, and give one economic reason for the difference.
8. Describe how the Bai-Perron test works.
9. Distinguish strict from weak exogeneity. Give a macroeconomic example where strict exogeneity clearly fails, and say what is lost.

10. A regressor is correlated with the error. Under what condition is the coefficient on the other regressor still unbiased?
11. With exogenous regressors and AR(1) errors, OLS is unbiased. What exactly is the problem, then?
12. Trace the four-step argument showing why a lagged dependent variable plus AR(1) errors produces bias.
13. Which failures leave the coefficients unbiased and which do not?
14. State the null hypothesis of the Breusch-Godfrey test and explain what the auxiliary regression is doing.
15. Why does the heteroskedasticity Q-test use squared residuals rather than residuals?
16. BPG gives p = 0.2285 and White gives p = 0.0925. Do these agree? What would you conclude at the 10% level?
17. Model 1 fails the normality test and Model 3 passes it. What does this suggest about the source of non-normality?
18. You have autocorrelated residuals and no lagged dependent variable. Give three possible remedies and say when each is appropriate.
19. Compare the ADL model with the AR(1)-error model. If they fit equally well, how would you choose, and what does each imply for monetary policy?
20. Why will HAC standard errors not rescue a model with a lagged dependent variable and autocorrelated errors?

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 10, 11 and 12 cover time series regression and serial correlation.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 7 covers autocorrelation.

Macroeconometric Forecasting. IMF online course. URL: https://www.edx.org/course/macroeconometric-forecasting-2.